

Week 4

Tutorial 4 — Block Diagrams, Signal Flow Graphs and Mason's Rule

QUESTION SHEET

This hour is reduction practice, and then the same diagram taken the other way. Question 1 is a nested two-loop reduction; Question 2 is an *overlapping*-loop diagram that needs a pick-off point moved before it will collapse; Question 3 converts that same diagram to a signal flow graph and evaluates it with Mason's rule, so you finish the hour having done one system by both routes and seen the answers agree. The homework extends Mason to a graph with a non-touching pair of loops, and revisits the canonical form with an unknown gain.

Questions marked **[in the hour]** are the ones we work through together. Those marked **[homework]** you do in your own time — they are not collected, but the technique is assumed from here on, and Assignment 2 (issued this week, due Week 6) covers exactly this material.

Preparation: Week 4 lecture notes, §1–§7.

Reading

Quick reference — carry these into every question below.

The three connections

cascade: G_1G_2 , parallel: $G_1 \pm G_2$

negative feedback: $\frac{G}{1+GH}$

Canonical form

$$T = \frac{C}{R} = \frac{G}{1+GH}, \quad \frac{E}{R} = \frac{1}{1+GH}$$

characteristic equation: $1+GH=0$

Moving a junction past a block G

summing junction, against the flow: $\div G$

pick-off point, with the flow: $\times \frac{1}{G}$

Mason's gain formula

$$T = \frac{\sum_k P_k \Delta_k}{\Delta}$$

$$\Delta = 1 - \sum_i L_i + \sum_{\text{non-touching pairs}} L_i L_j - \dots$$

P_k : gain of the k -th forward path.

L_i : gain of the i -th loop, signs included.

Δ_k : Δ with every loop touching path k deleted.

Non-touching = sharing *no node*.

Second-order denominator

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Question 1 — Nested loops: reduce from the inside out *[in the hour]*

Figure 1 shows a position servo with an inner rate loop. The amplifier has gain $G_1 = 10$, the power stage is $G_2(s) = 1/(s+5)$, the shaft integrates its input so $G_3(s) = 1/s$, the rate sensor has gain $H_1 = 3$, and the position sensor is a unit-gain potentiometer.

- Collapse the inner loop to a single block, and say what feedback has done to the pole of G_3 .
- Cascade the amplifier and power stage with your result, then collapse the outer loop to find $T(s) = C(s)/R(s)$.
- Identify ω_n and ζ , and give the closed-loop poles.

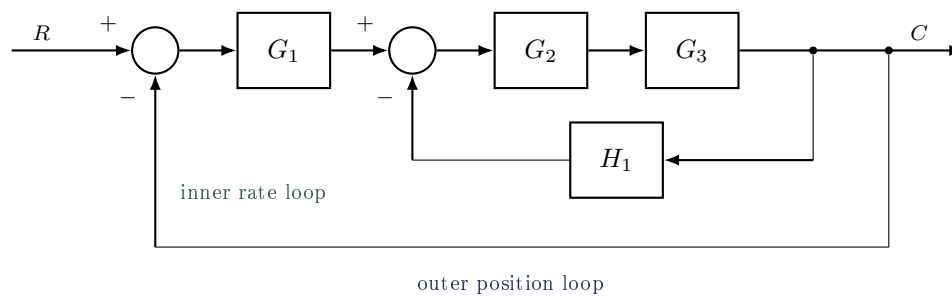


Figure 1: Question 1: a position servo with an inner rate loop.

- (d) What is the d.c. gain of the closed loop? Why is it not 1, given that the forward path contains an integrator?

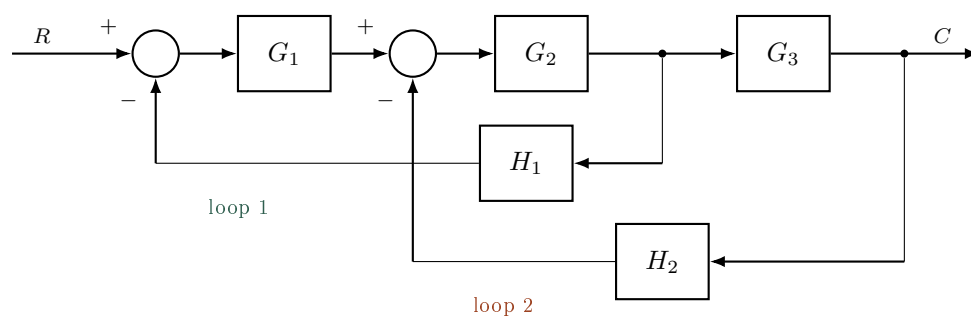
Common pitfall

It is tempting to write “the forward path has an integrator, so the d.c. gain is 1”. That rule belongs to the *outer* loop’s forward path, and in this diagram the integrator is inside the inner loop, where feedback has already relocated its pole. Always check which loop a block ends up inside after the first collapse.

Question 2 — Overlapping loops: move a pick-off point *[in the hour]*

Figure 2 has two loops that overlap: loop 1 (through H_1) spans G_1 and G_2 , and loop 2 (through H_2) spans G_2 and G_3 . The values are

$$G_1 = 2, \quad G_2(s) = \frac{1}{s+1}, \quad G_3(s) = \frac{4}{s+6}, \quad H_1 = 1, \quad H_2 = 0.5.$$

Figure 2: Question 2: overlapping loops. Loop 1 spans G_1, G_2 and loop 2 spans G_2, G_3 — neither contains the other.

- Explain in one sentence why this diagram cannot be reduced by collapsing an inner loop first.
- Move loop 1’s pick-off point forward through G_3 , stating the gain the moved branch must now carry.
- Reduce the resulting nested diagram to find $T(s) = C(s)/R(s)$, in factored form.
- State the closed-loop poles and the d.c. gain.

Question 3 — The same system, by SFG and Mason [in the hour]

Now take the *original* diagram of Figure 2 — before any junction was moved — and evaluate it the other way.

- Draw the signal flow graph. Use nodes R , E_1 (the output of the first summing junction), E_2 (the output of the second), X (the output of G_2) and C .
- List every forward path with its gain, and every loop with its loop gain. Are the two loops touching or non-touching? Justify your answer by listing each loop's nodes.
- Write down Δ and the cofactor Δ_1 , and evaluate Mason's formula.
- Compare with your answer to Question 2(c). What have you gained, and what have you lost, by using Mason's rule instead of reduction?

Key idea

Doing one system both ways, as Questions 2 and 3 just did, is the best available check on a reduction. If the two routes disagree, the error is almost always a lost sign at a summing junction or a forgotten compensating factor on a moved branch — look there first, not at the arithmetic.

Question 4 — Mason with a non-touching pair [homework]

Figure 3 is a signal flow graph with a feedforward branch. Take

$$a_1 = 2, \quad a_2(s) = \frac{1}{s+1}, \quad a_3(s) = \frac{4}{s+2}, \quad a_4 = 1, \quad a_5 = 0.5, \quad b_1 = 1, \quad b_2 = 1.$$

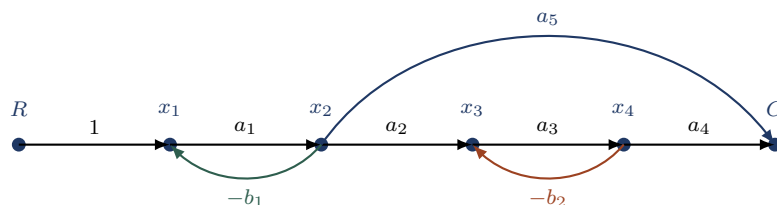


Figure 3: Question 4: a graph with two forward paths and two loops.

- Identify both forward paths and both loops, with their gains.
- Show that the two loops are non-touching, and hence write down Δ .
- Show that your Δ factors as $(1 - L_1)(1 - L_2)$. Explain why this factorisation is guaranteed whenever the graph's only two loops are non-touching.
- Find both cofactors Δ_1 and Δ_2 , and hence $T(s) = C/R$.
- State the closed-loop poles and the d.c. gain. Where did the numerator's zeros come from — and would they be there if a_5 were removed?

Common pitfall

The commonest error in this question is writing $\Delta_2 = 1$ out of habit. A cofactor is 1 only when the path touches *every* loop. Before writing Δ_k , list the nodes on path k and strike out only the loops that share one — here that is loop 1 alone, which is why loop 2 stays in Δ_2 .

Question 5 — The canonical form with an unknown gain *[homework]*

A unity-feedback system has forward path

$$G(s) = \frac{K}{s(s+5)}, \quad K > 0.$$

- Write down the open-loop transfer function, the characteristic equation and the closed-loop transfer function $T(s)$ in terms of K .
- Find the value of K that gives a natural frequency $\omega_n = 4 \text{ rad s}^{-1}$, and the damping ratio ζ that comes with it.
- Give the closed-loop poles for that K .
- Find the steady-state error to a unit step, using $E(s)/R(s)$. Explain the answer in one sentence.
- If K is raised to 64, what happens to ω_n and ζ ? State the general relationship between K and ζ for this system, and say what it implies about raising gain to make a system faster.

Question 6 — Two inputs, one characteristic equation *[homework]*

In Figure 4 the reference is R , and U is a second input entering between the two forward blocks. Take $G_a = 5$, $G_b(s) = 1/(s+3)$ and unity feedback.

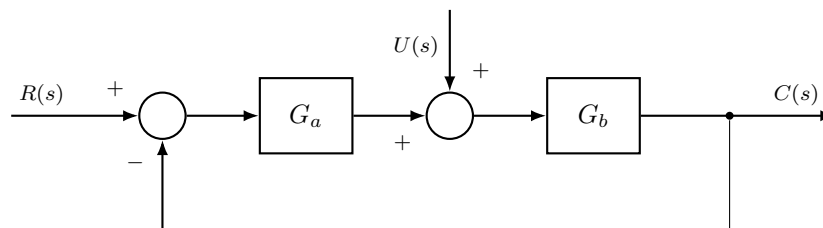


Figure 4: Question 6: a system with a second input entering after G_a .

- Find $C(s)/R(s)$ with $U = 0$, and $C(s)/U(s)$ with $R = 0$.
- Write the total response $C(s)$ in terms of both inputs.
- Why do the two transfer functions have the same denominator?
- R is a unit step and U is a constant -2 (a steady load pulling the output down). Find the steady-state value of $c(t)$.
- What constant value of U would hold the output at zero in the steady state while the unit step at R is still applied?

Solutions to the homework questions are released with the Week 5 sheet.

Next week: transient response of first- and second-order systems.